

Replication Summary: PRISM — Unlocking the Black Box of CNNs

Replication of Szandała, T. (2023). *Unlocking the black box of CNNs: Visualising the decision-making process with PRISM*. Information Sciences, 642, 119162.

Objective

The original paper introduces **PRISM (Principal Image Sections Mapping)**, an explainable-AI (XAI) method that applies Principal Component Analysis (PCA) to the internal feature representations of a CNN, processed as a *batch*, to produce an RGB-coloured overlay highlighting features the model uses to distinguish between images. Unlike single-image saliency methods (e.g., Grad-CAM), PRISM is designed to expose *comparative*, distinguishing features across multiple images at once. This project reproduces three of the paper’s central experimental results using an independent implementation.

Method

A pretrained **VGG-16** (ImageNet-1k weights) was used with a forward hook attached to its last convolutional layer, yielding a (batch, 512, 14, 14) feature representation. PRISM’s core algorithm (reshape → centre → SVD → keep top-3 principal components → reshape to RGB) was re-implemented from the paper’s Algorithm 1 pseudocode, independently of the authors’ public `TorchPRISM` repository. Test images were sourced from ImageNet for classes matching the paper’s examples (timber wolf, coyote, grey fox, border collie, samoyed, electric locomotive, train cars). Bicubic upsampling was used in place of the paper’s Gradual Extrapolation technique to render overlays. Random seed 42 was fixed for reproducibility.

Experiments and Results

1. Core PRISM Visualisation (Sec. 3.3, Fig. 8)

On a batch of two wolf images and one coyote image, PRISM produced a colour overlay concentrated on the animals’ muzzles, with background regions (dirt, rock) receiving minimal colour — matching the paper’s finding that the muzzle is the primary distinguishing feature between wolves and coyotes.

2. PCA Variance Distribution (Sec. 5.2, Figs. 16–18)

The fraction of variance captured by PRISM’s top-3 principal components was compared across a 2-class batch (3 images: wolf, coyote) and a 5-class batch (10 images: wolf, coyote, grey fox, border collie, samoyed):

Batch	Images	Classes	Top-3 PC Variance
Small	3	2	47.4%
Large	10	5	32.6%

The paper reports >57% variance for a comparable 5-class batch and states that variance captured decreases as class diversity increases. Our results reproduce this *trend* exactly (47.4% → 32.6%); absolute values differ due to different source images, a shallower hooked layer (14 × 14 vs. the paper’s 7 × 7), and fewer images per class.

3. PRISM vs. Grad-CAM Comparison (Sec. 3.2/4.1, Figs. 6, 7, 11)

On a wolf/wolf/grey-fox set, Grad-CAM produced structurally similar heatmaps (a hotspot on the head) for all three images, while PRISM produced visibly different colouring between the wolves and the fox — reproducing the paper’s claim that PRISM exposes distinguishing features that saliency maps cannot. On a second set (electric locomotive / train cars), the model misclassified the train-cars image as “*harvester*”; Grad-CAM still localised correctly onto the train, while PRISM produced a flat, uninformative overlay. This reproduces a documented limitation of PRISM (Sec. 4.2): it is *target-agnostic* and does not filter for relevance to the model’s final predicted class.

Conclusion

All three experiments qualitatively reproduce the corresponding claims in the original paper: PRISM correctly localises distinguishing features (Experiment 1), its top-3-PC variance efficiency decreases with batch diversity as claimed (Experiment 2), and it differentiates features in a way single-target saliency maps cannot, while also exhibiting the paper’s documented target-agnosticism limitation (Experiment 3). Absolute numerical values differ from the paper due to differing source images, hooked-layer resolution, and omission of Gradual Extrapolation; these differences are documented and do not affect the validity of the underlying trends. Representation-swapping (Sec. 3.1), the 9-model applicability test (Sec. 3.3), SOM clustering (Sec. 5), and Vision Transformer analysis (Sec. 5.1) were outside this project’s scope. Full code, notebooks, and a detailed comparison are available in the accompanying GitHub repository.